DIGITIZER TEST REPORT

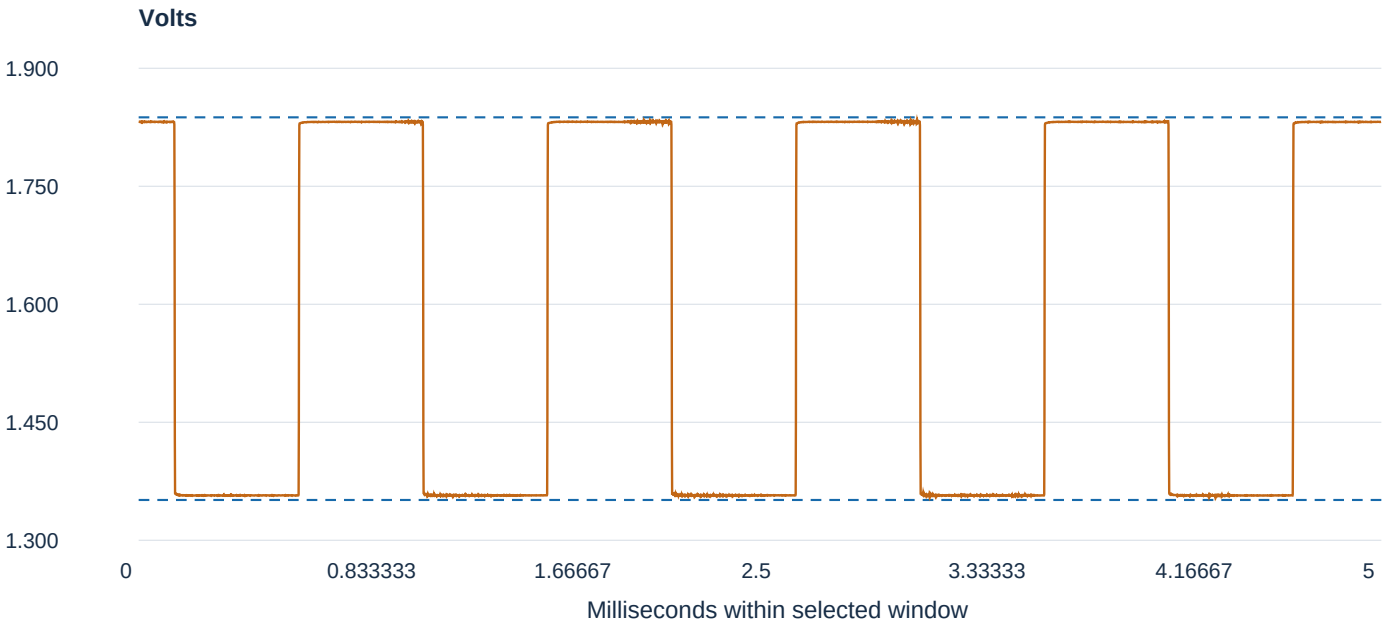
1 kHz square-wave recording

Rigol external source | ADS127 at 1,024,000 samples/s | 30 October 2026

PASS - recording, readback and waveform checks

30.000 seconds recorded; no reported receiver errors or recording-buffer overflow.
92,159,133 sample bytes; one file changeover; retained-file checksums matched.

Five cycles, shown sample by sample



Orange: 5,120 ADC samples. Blue dashed: entered scope reference levels.
This 5 ms window begins 1.000 s into the recording. It is not a 30-second average.

Level	Reference (V)	ADC mean (V)	Difference (mV)	Std. dev. (uV)
Vbase / low	1.3513	1.357112	+5.812	208.87
Vtop / high	1.8377	1.831946	-5.754	126.90

Recorded frequency: 1000.00 Hz. ADC high-to-low swing: 474.834 mV.

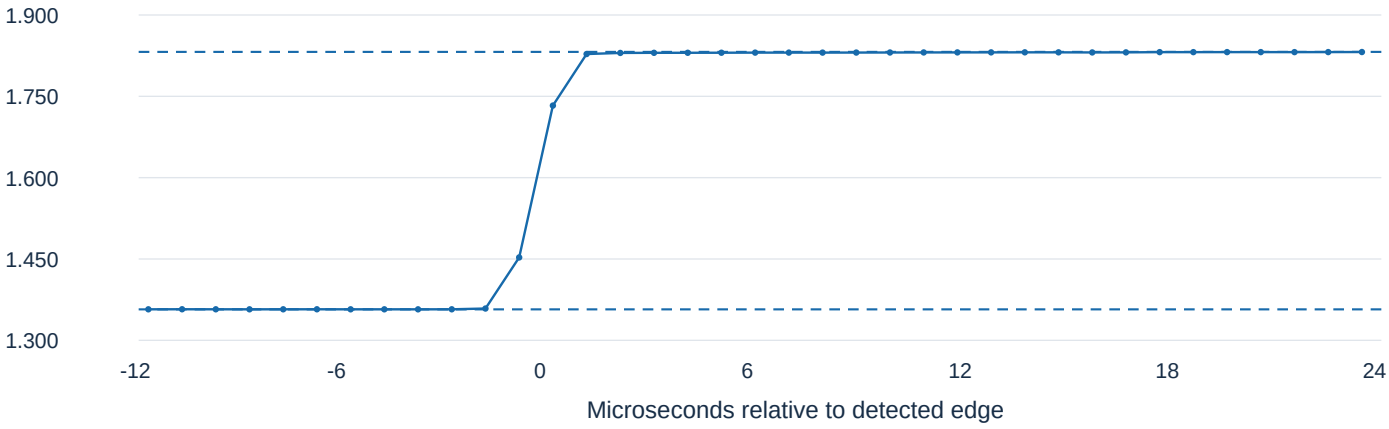
What these measurements establish

The recorded signal has the expected 1 kHz frequency and approximately 50% duty cycle.
Means and standard deviations use the middle half of complete high/low intervals:
1,280 low-level samples and 1,024 high-level samples, all within this 5 ms window.
The reference entries (1.3513 / 1.8377 V) match earlier scope readings. Their validity
at 1 kHz has not been confirmed; voltage differences are provisional.
ADC volts assume a 2.5 V reference. Frequency uses the configured ADC sample rate.
This is a functional test, not an independent calibration of voltage or time.

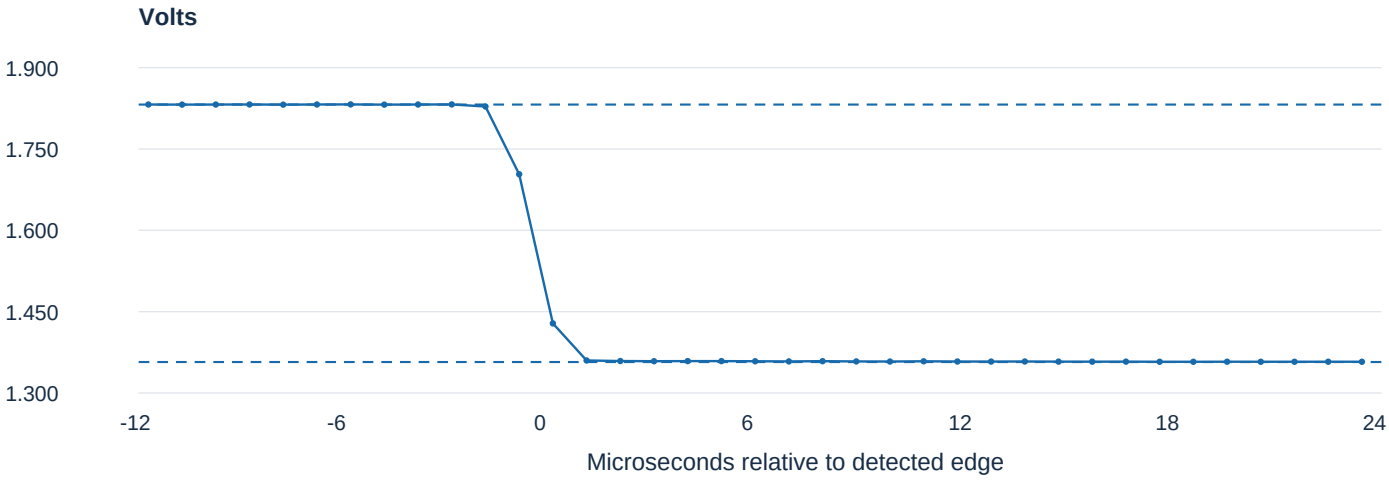
A closer look at the edges

Individual samples around a rising and a falling transition

Rising edge



Falling edge



Dots are individual samples, 0.977 us apart; connecting lines are only a visual guide.
Dashed lines mark the measured settled levels. Zero is detection, not generator launch.

Measurement across complete edge intervals	Result
Largest excursion above settled high level	0.861 mV
Largest excursion below settled low level	0.996 mV
Latest settling after edge detection	0.977 us (one sample)
Settling band around destination level	+/-9.497 mV (2% of ADC swing)

Settling means samples enter and remain in that band, excluding the end of the interval.
One-sample settling does not measure a precise analog response time. Excursions include noise and do not by themselves prove ringing. Source and ADC contributions are combined.